

# Standard Model Particles from GF(27)\* — all quantum numbers algebraically forced [B]

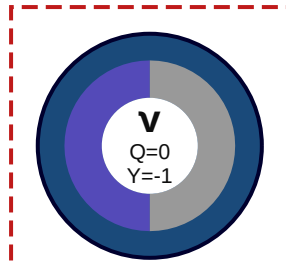
FFGFT · J. Pascher 2026 · Doc. 346/347/348/349

Generation 1

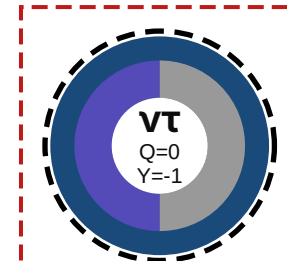
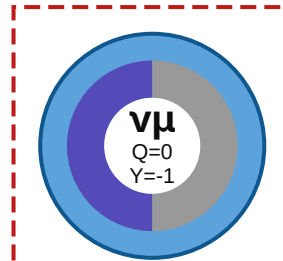
Generation 2

Generation 3

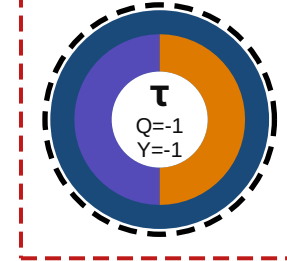
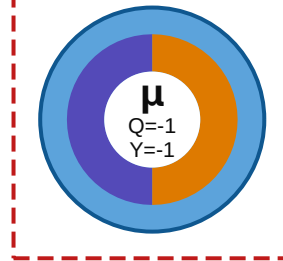
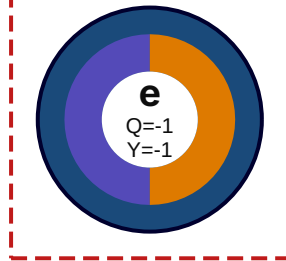
Neutrinos  
(QR)



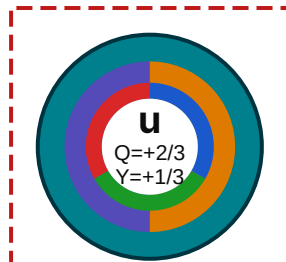
SU(2)L-doublet = Sd-sector [B] Doc.349



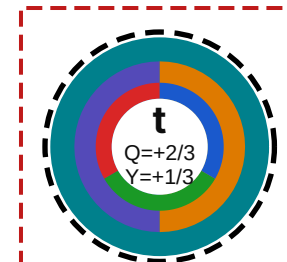
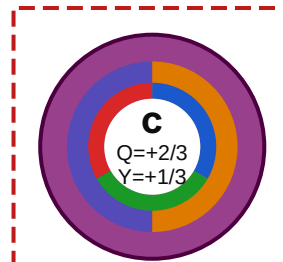
Charged  
leptons  
(QR)



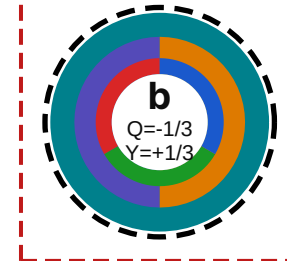
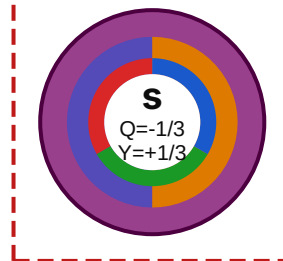
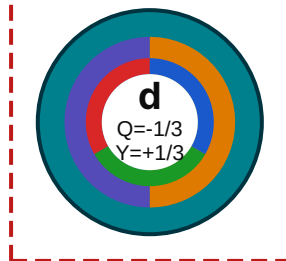
Up quarks  
(NQR)



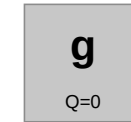
SU(2)L-doublet = Sd-sector [B] Doc.349



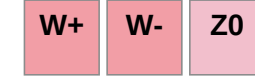
Down  
quarks  
(NQR)



## Gauge bosons



GF(3) fixed field [B] Doc.339



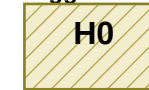
Gal(GF(9)/GF(3))=Z2 [K] Doc.336

## Gluons gx8



4 Z13-Orbits x 2 Z2-par. = 8 [B]

## Higgs



Q=0

T~.m=1 replaces Higgs (Doc.019) Exp. established [X]

## Legend

Outer ring = Galois orbit:

- O1/O3: Gen.1/3 leptons (blue)
- O3: Gen.2 leptons (light blue)
- O2: Gen.1/3 quarks (teal)
- O4: Gen.2 quarks (berry)

Middle ring = chirality:

- Left violet = Sd = left-h. [B]
- Right orange = Su = right-h. [B]
- Gray = vR absent; Dirac [B/S]

Colour ring (quarks only):

- r/g/b = colour charge [B] Doc.346

Red dashed = SU(2)L doublet

Black dashed = Gen.3

[B] algebraically proved

[K] numerically [X] experimentally